

Project 4 (C++): Given a text file of English words, your task is to compute character frequencies (counts) in the input file, then output the result.

** You will be given two data to test your program: data1 and data2; data1 is very simple, you can eyeball the counts.

1) Implement your program according to specs below until it passes the compilation.

2) Run your program with data1 and eyeball the correctness of your program.

3) Run your program with data2

*** Include in your hard copy

- Cover page
- Source code
- data1 // with caption, "**** below is data1****"
- outFile for data1 // with caption, "**** below is outFile for data1"
- logFile for data1 // with caption, "**** below is logFile for data1" // Limited to 3 pages if more.
- data2 // with caption, "**** below is data2****"
- outFile for data2 // with caption, "**** below is outFile for data2"
- logFile for data2 // with caption, "**** below is logFile for data2" // Limited to 3 pages if more.

Note: You must use argv to open input and output files **-2 points deduction if you hard-code your file names!**

Language: C++

Project name: Compute frequency of characters in a file

Project points: 10 pts Compute frequency of characters in a file

Due Date: Soft copy (*.zip) and hard copies (*.pdf):

(10/10 pts): on time, 10/9/2026 Friday before midnight (11:59pm).

(-10/10 pts): non-submission, 10/9/2029 Friday after midnight

*** All submission MUST include soft copy (*.zip) and hard copy (*.pdf) in the same email attachments with correct email subject as stated in the project submission requirement; otherwise, your submission will be rejected. -10/10.

Email subject: (313) first name last name <Project 4: Compute frequency of characters in a file (C++)>

*** Inside the email body includes:

- Your answer to the five questions given at the end of the project submission requirements. (-1 if omitted.)
- Screen recoding link. (-2 if omit screen recording).

*** Place your screen recording in your project submission email body below the 5 questions.

I. Input (argv [1]): A text file contains English language.

II. Outputs: a) outFile (argv [2]): As dictate by the program.

b) logFile (argv [3]): As dictate by the program.

III. Data structure:

- counting class:

- (int) charCountAry [256] // a 1-D array to store the character counts; initialize to zero.

Methods:

- constructor ()
- initZero (...) // initialize charCountAry to zero. On your own.
//If you like, you may place this method inside constructor.
- computeCharCounts (...) // See algorithm below.

- printCountAry (...oFile) // print the character count array to oFile (oFile may be outFile or logFile)
 //in the following format:

Ascii	character	count
1	char	count
2	char	count
:	:	:
:	:	:

**** >>> **Print only character with count > 0**

**** >>> **Do not print ascii 13 or 23, these will cause printing problem.**

**** >>> The ascii code for blank is 32; the ascii code for \n (new line) is 10; therefore, do not
 Convert index 10 to char and do not convert index 32 to char, instead, for
 index 10 and 32, print as follows:

10	\n	count
32	blank	count

For example:

Ascii	character	count
10	\n	52
32	blank	87
49	1	4
65	A	42
66	B	38
67	C	122
:	:	:
:	:	:

IV. main (...)

Step 0: //check argc count is correct and check each file if it can be opened.

inFile, outFile, logFile ← open from argv[1], argv[2], argv[3]

Step 1: initZero (charCountAry)

Step 2: computeCharCounts (inFile, charCountAry, logFile)

Step 3: outFile ← “**** In main printing charCountAry”

printCountAry (charCountAry, outFile)

Step 4: close all files.

V. computeCharCounts (inFile, charCountAry, logFile)

Step 0: logFile ← “**** Entering computeCharCounts method”

Step 1: ch ← read a character from inFile

Step 2: index ← (int) ch

Step 3: charCountAry[index]++

Step 4: logFile ← “**** In computeCharCounts () ch = index = charCountAry[index] = ” // print all 3 values.

Step 5: repeat Step 1 to Step 4 until inFile is empty

Step 6: logFile ← “**** In computeCharCounts method, print the charCountAry ****”

printCountAry (charCountAry, logFile)

Step 7: logFile ← “**** Leaving computeCharCounts () method”